Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-IL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

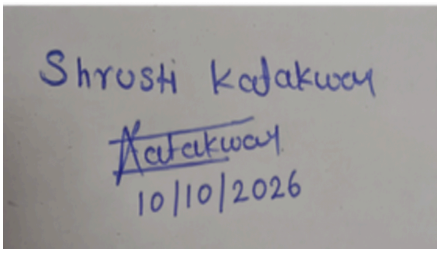
Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)



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Session	**2026-27 (ODD)**	**Course Name**	**HCPA**
Semester	**5th**	**Course Code**	**23ADS1524**
Roll No	**119**	**Name of Student**	**Shrusti Katakwar**

Practical Number	**07**
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	Implement matrix multiplication in parallel using OpenMP or CUDA.
Problem Definition	Implement matrix multiplication in parallel using OpenMP or CUDA.
Theory (100 words)	Matrix multiplication is a mathematical operation in which two matrices are multiplied to produce a third matrix. In parallel matrix multiplication, multiple calculations are performed simultaneously to reduce execution time. Each element of the result matrix is calculated by multiplying the elements of a row of the first matrix with the corresponding elements of a column of the second matrix and adding the products. OpenMP uses multiple CPU threads to perform calculations in parallel, whereas CUDA uses GPU threads for parallel processing. This technique improves performance and is widely used in scientific computing, artificial intelligence, machine learning, and image processing.

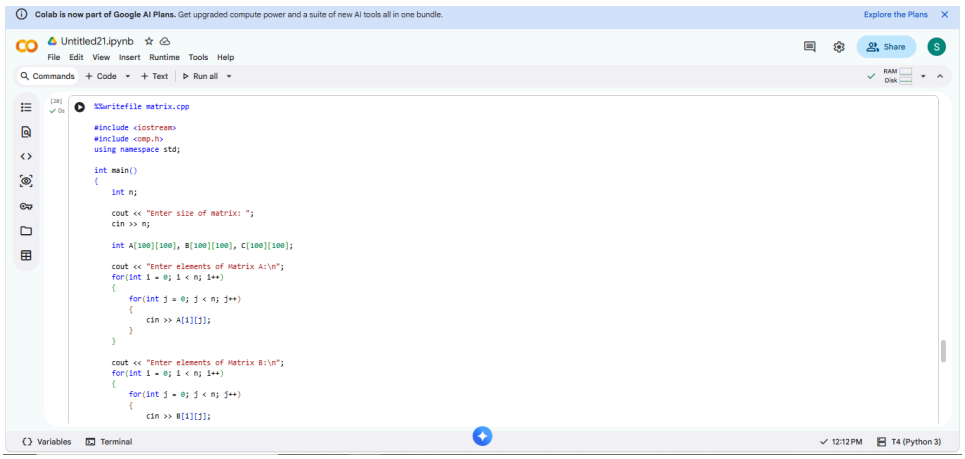
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Procedure and Execution (100 Words)	Step for Implementation: 1.Initialize Matrices: Create two input matrices A and B with suitable dimensions. 2. Input Values: Assign values to both matrices. 3. Allocate Result Matrix: Create a result matrix C to store the multiplication output. 4. Implement Parallelism: Use OpenMP to divide calculations among multiple CPU threads or CUDA to execute them on GPU threads. 5. Perform Multiplication: Each thread calculates one or more elements of the result matrix. 6. Store Results: Store the calculated values in the corresponding positions of matrix C. 7. Display Output: Print the resulting matrix C. 8. Verify Results: Compare the output with sequential matrix multiplication to check correctness.

	Code:
	 
	Colab is now part of Google AI Plans. Get upgraded compute power and a suite of new AI tools all in one bundle. Explore the Plans
	Untitled21.ipynb
	File Edit View Insert Runtime Tools Help
	Commands + Code + Text Run all
	%%writefile matrix.cpp
	#include <iostream>
	#include <omp.h>
	using namespace std;
	int main()
	{
	int n;
	cout << "Enter size of matrix: ";
	cin >> n;
	int A[100][100], B[100][100], C[100][100];
	cout << "Enter elements of Matrix A:\n";
	for(int i = 0; i < n; i++)
	{
	for(int j = 0; j < n; j++)
	{
	cin >> A[i][j];
	}
	}
	cout << "Enter elements of Matrix B:\n";
	for(int i = 0; i < n; i++)
	{
	for(int j = 0; j < n; j++)
	{
	cin >> B[i][j];
	Variables Terminal
	12:12 PM T4 (Python 3)

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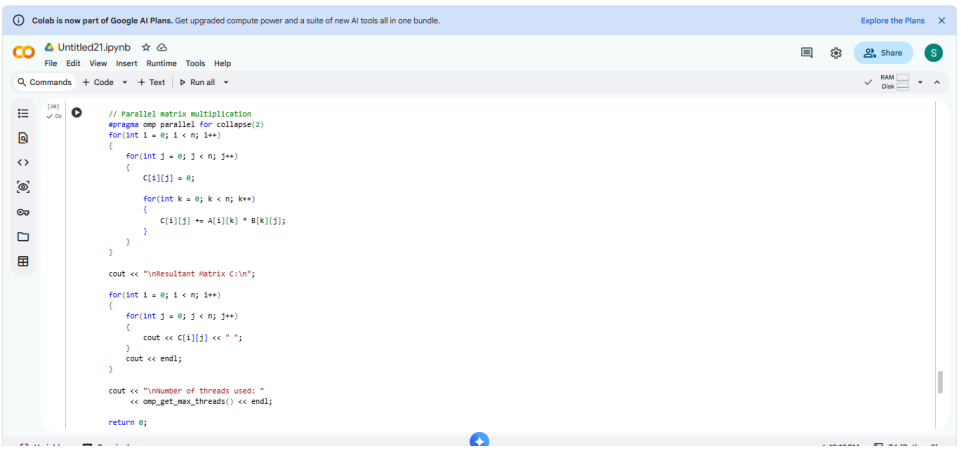
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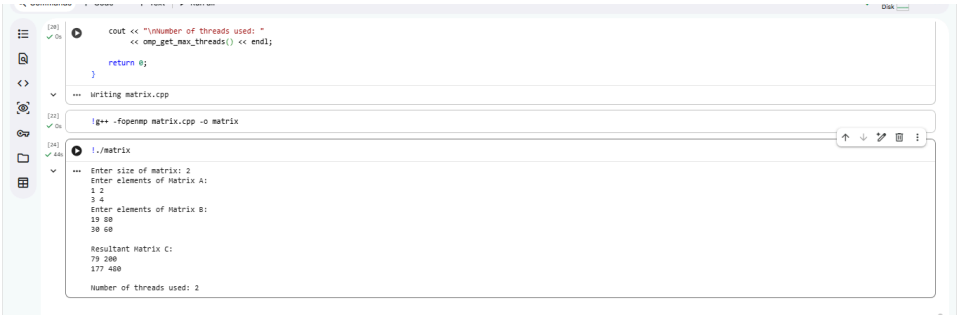
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Output:



Output Analysis

1. The program successfully multiplies two input matrices and displays the result matrix.
2. Parallel processing distributes calculations among multiple threads.
3. Each thread calculates different elements of the result matrix simultaneously.
4. OpenMP uses CPU threads, while CUDA uses GPU threads for computation.
5. Parallel execution reduces computation time for large matrices compared to sequential execution.
6. The output is verified by comparing it with the result of sequential matrix multiplication.

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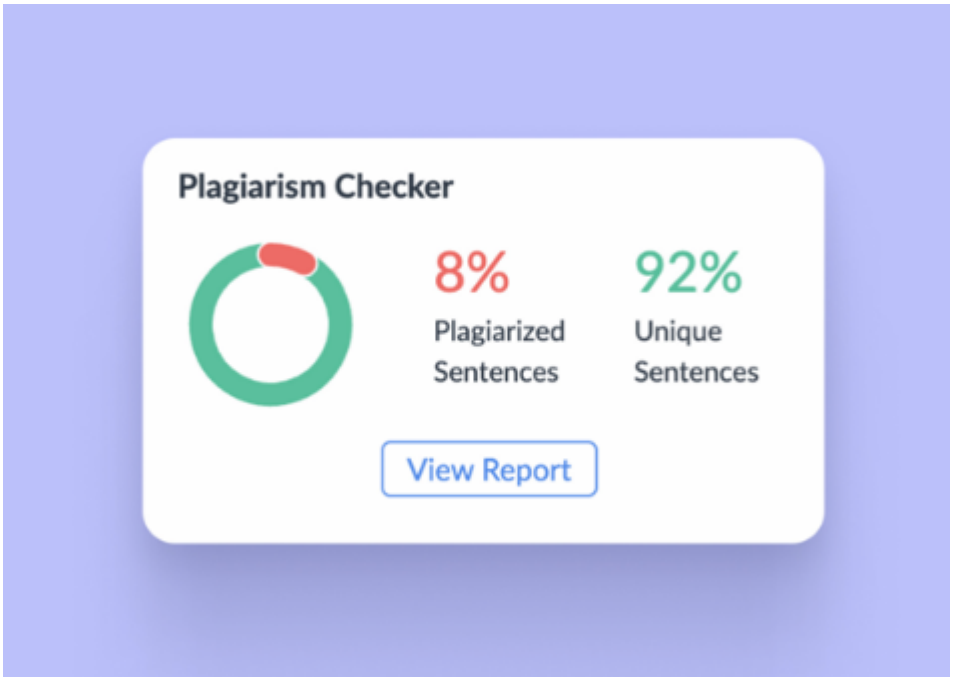
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Link of student Github profile where lab assignment has been uploaded	**https://github.com/Shrusti-Katakwar**
Conclusion	Parallel matrix multiplication was successfully implemented using OpenMP or CUDA. The program produced the correct result by performing multiple calculations simultaneously. Parallel processing reduced execution time, especially for large matrices, compared to sequential processing. Thus, parallel computing improves computational efficiency and is useful for scientific applications, artificial intelligence, and machine learning.
Plag Report (Similarity index < 12%)	
Date	**10/10/2026**